

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458444.6005 +/- 0.0033 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.207 +/- 0.023
Transit depth (Rp/Rs)^2	4.3 +/- 0.94 [%]
Orbital Inclination (inc)	85.61 +/- 1.56
Airmass coefficient 1 (a1)	1.017 +/- 0.019
Airmass coefficient 2 (a2)	-0.01 +/- 0.17
Scatter in the residuals of the lightcurve fit is	1.86 %
Best Comparison Star	#2 - [414, 336]
Optimal Aperture	3.15
Optimal Annulus	9.97
Transit Duration (day)	0.0735 +/- 0.0077

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.207 ± 0.023 vs 0.1646 (deviation 1.33σ)
Duration fit vs published	0.0735 d vs 0.0758 d (deviation 0.22σ)
Mid-time O–C	0.5 min
Depth SNR	6.5
Residual scatter	1.86 %

Light curve

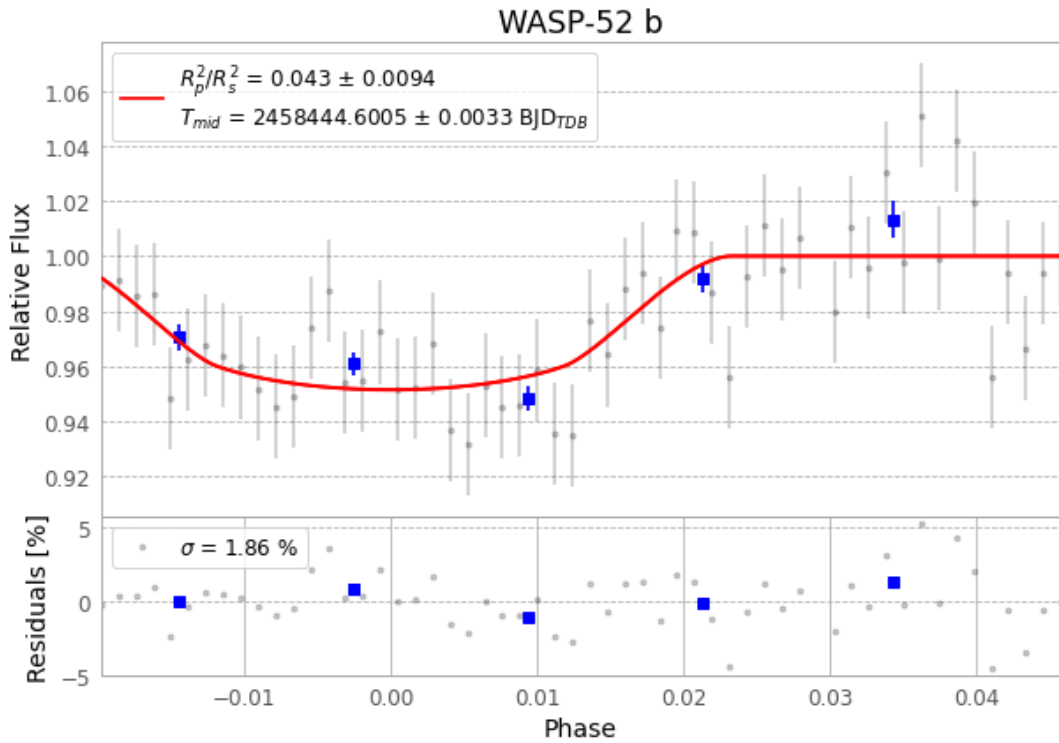


Figure 1 — FinalLightCurve_WASP-52 b_2018-11-21.png (SHA-256 6269ba877ffe12e7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [294, 253], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ac5f13035e4c28af...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

56 FITS frames (37 MB), WASP-52181122013012.FITS → WASP-52181122041512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-181122.log (71bfc0fa3b13...), FinalLightCurve_WASP-52 b_2018-11-21.png (6269ba877ffe...), FinalLightCurve_WASP-52 b_2018-11-21.csv (d4ac020bf291...)

Compute resources

Wall time 100.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 06:08Z.